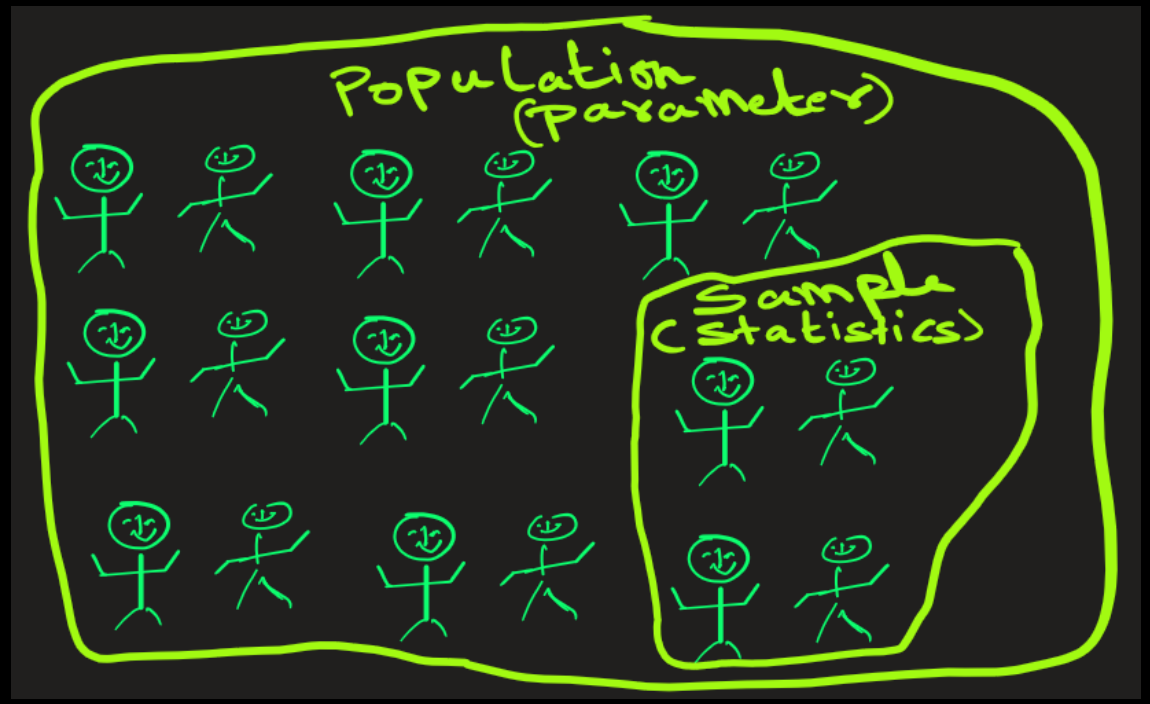
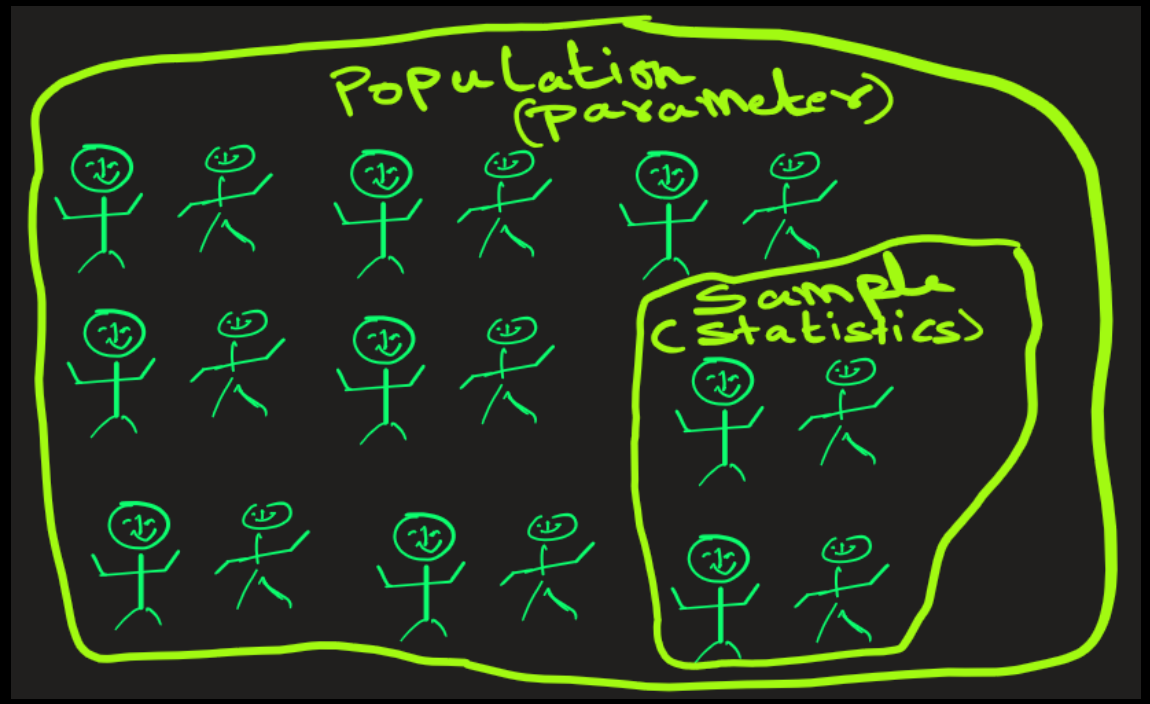
Population versus Sample

Concept	Definition	Example
Population	The **entire group** you want to study or make conclusions about.	All the students in your university.
Sample	A **subset of the population**, selected for actual observation and analysis. Used when studying the entire population is impractical.	200 students chosen randomly from your university.



Parameter versus Statistics

Concept	Definition	Symbol	Example
Parameter	A **numerical value** that describes a **characteristic of a population**. It is usually **unknown** and **fixed**.	Greek letters (μ, σ, p, etc.)	Population mean (μ) = average height of all students in the university.
Statistic	A **numerical value** that describes a **characteristic of a sample**. It is **computed from sample data** and used to **estimate** the population parameter.	Latin letters ($\bar{x}$, s, $\hat{p}$, etc.)	Sample mean ($\bar{x}$) = average height of the 200 students sampled.



stop